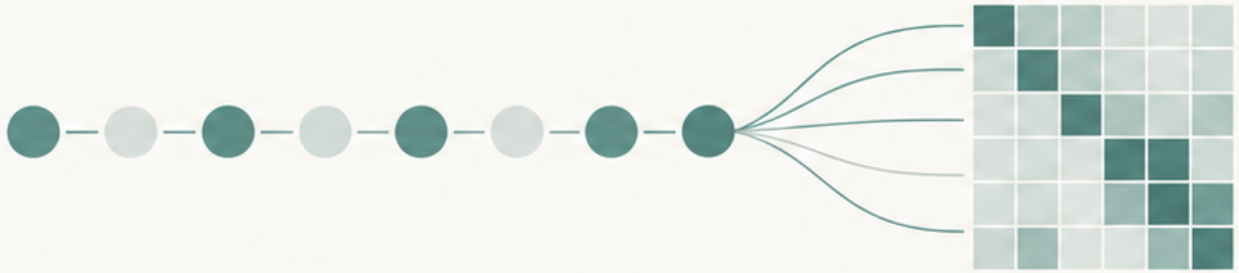
Transformers:
From Attention to Time Series Forecasting

Machine Learning Module
Master's in Data Science and Analytics

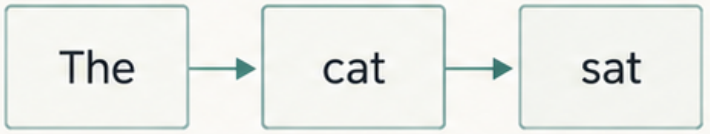


Camilo Solano David Ruiz Juan Otálvaro

Why Do We Need Better Sequence Models?

Sequence data

Text



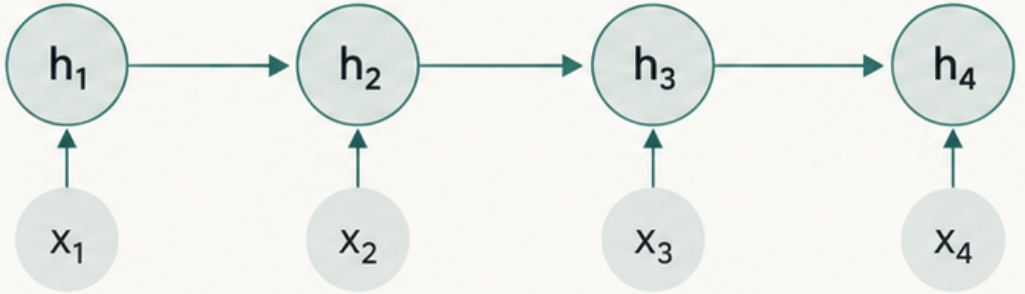
Time series

Events

What a model must capture

- Order matters
- Relevant context may be far apart
- Temporal patterns carry information

RNN / LSTM: step-by-step processing

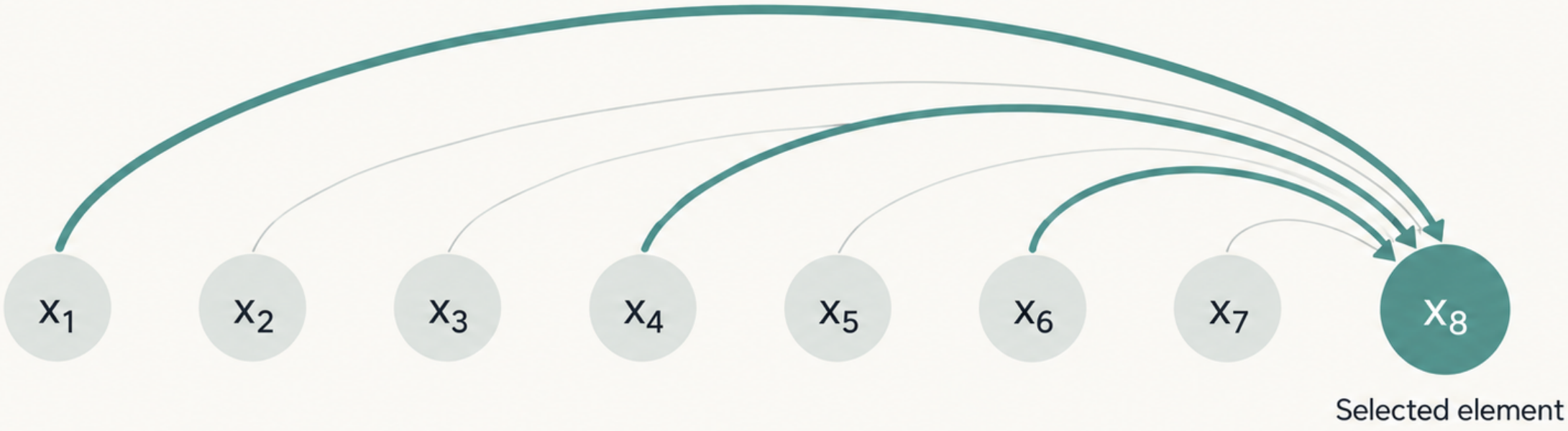


Long-range context can be hard to retain

Sequential computation limits parallel training

Attention: Focus on What Matters

Each element can learn which other elements are most relevant.



- Dynamic weighting
- Captures long-range relationships
- Not all past observations matter equally

Self-Attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

d_k: key dimension

Query (Q)

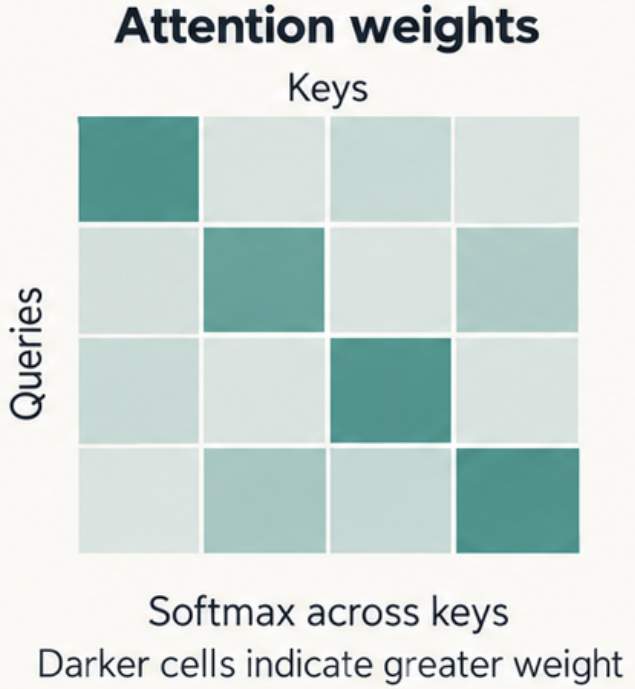
What this position is looking for

Key (K)

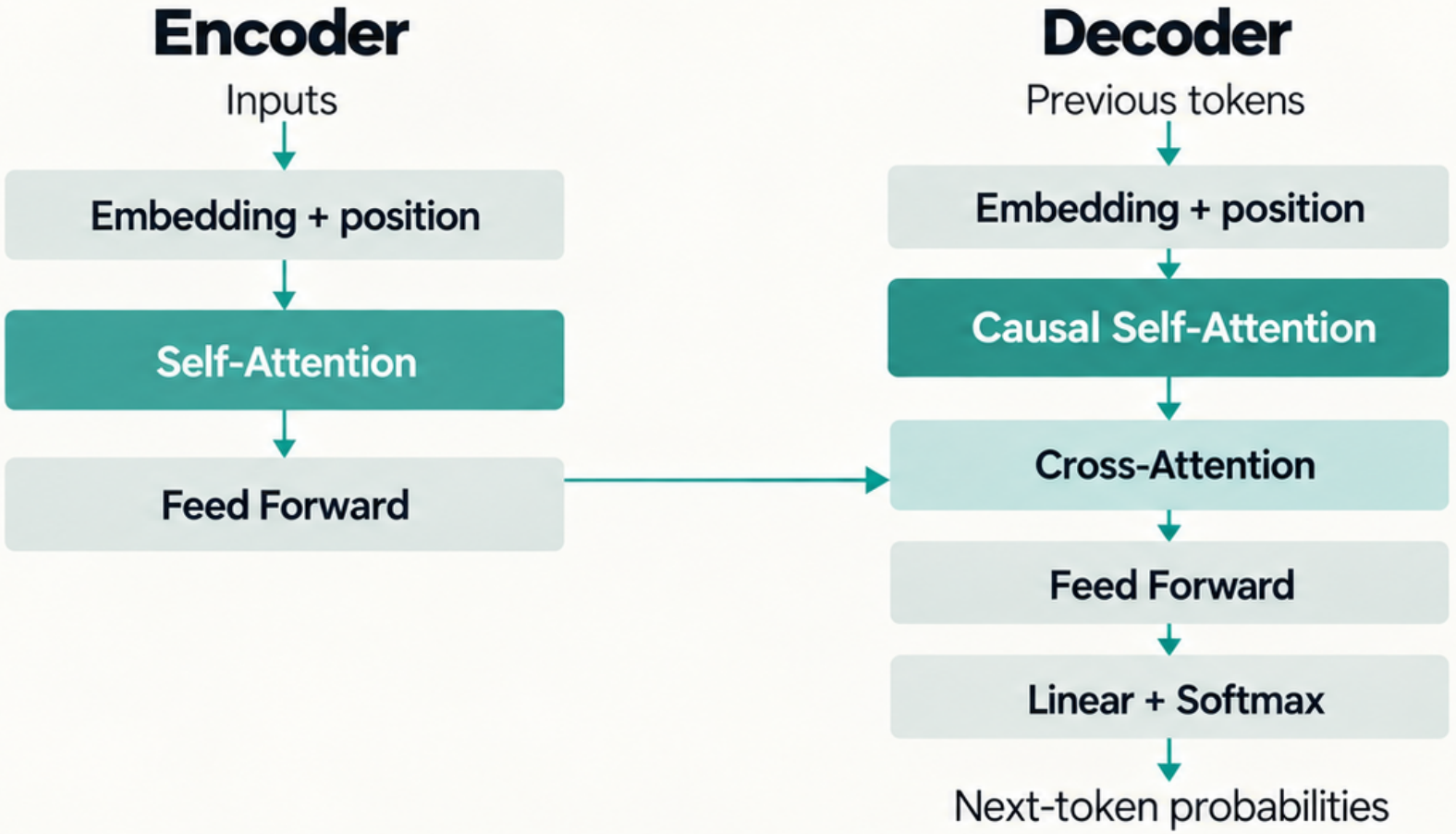
What each position offers

Value (V)

The information to combine



Transformer Architecture

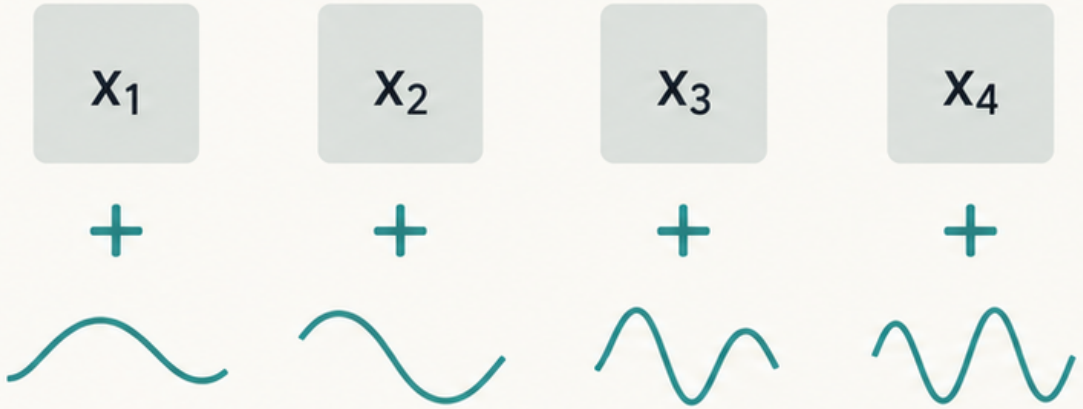


Encoder-only	Decoder-only	Encoder–Decoder
BERT-style	GPT-style	Original Transformer / T5
Representation / classification	Autoregressive generation	Sequence-to-sequence

Modern Transformer models often use only the encoder, only the decoder, or both.

Two Key Ideas: Position and Multiple Heads

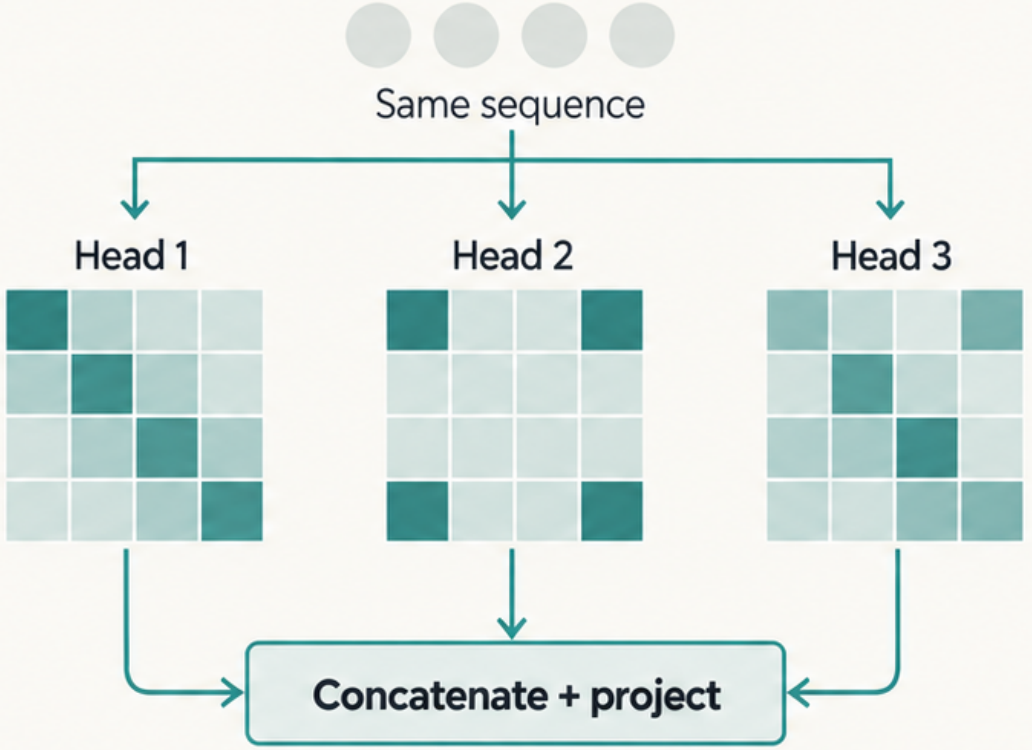
Positional encoding



Add a position signal to each token

Attention alone does not know order.

Multi-head attention



Different heads learn different relationships.

Why Transformers Matter

Strengths

Parallel training

Process sequence positions together

Long-range interactions

Connect distant elements directly

Flexible sequence representation

Model text, time series, or ordered events

Limitations

Quadratic attention cost

Standard attention cost grows quadratically with sequence length

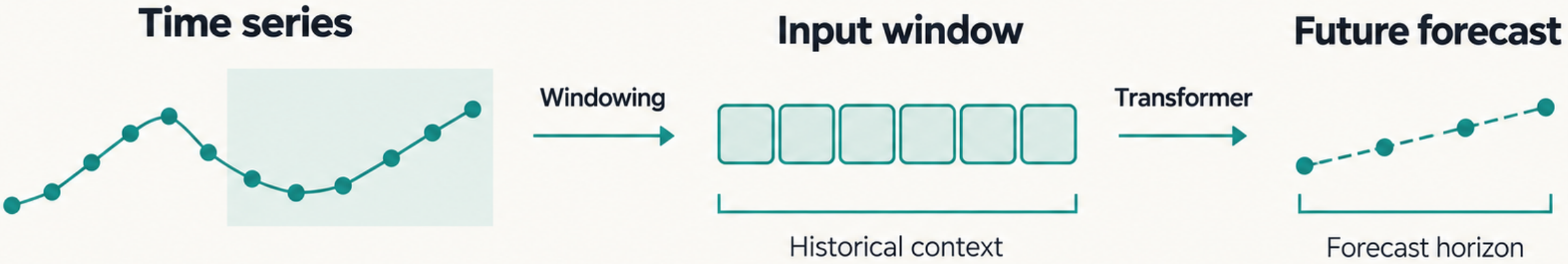
Data requirements

Training from scratch can require substantial data

Forecasting gains vary

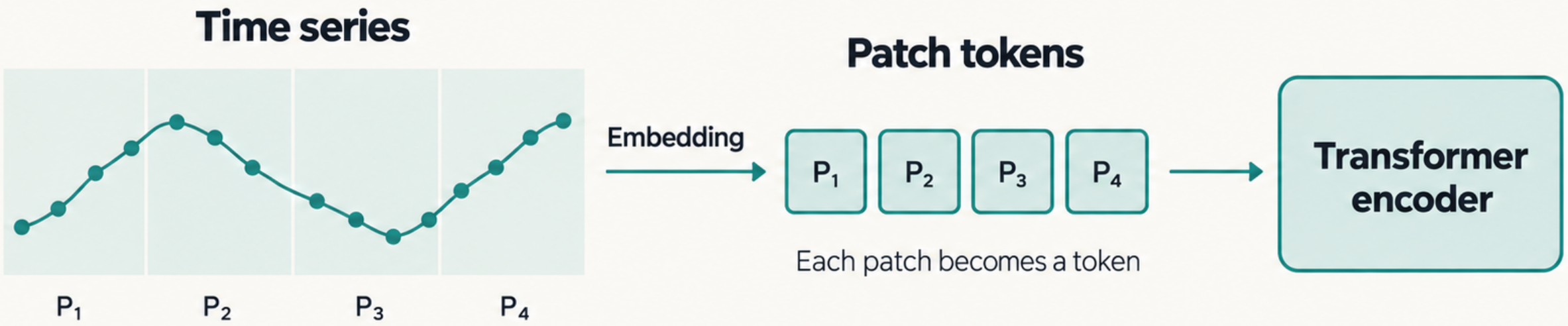
Simple baselines can still outperform

Transformers for Time Series



In time series, observations or patches become the sequence elements.

PatchTST: A Time Series Is Worth 64 Words



Patching reduces sequence length
Fewer tokens for attention

Longer context
Use more of the past

Channel-independent design
Shared weights across variables

Nie et al., ICLR 2023

Temporal Fusion Transformer (TFT)

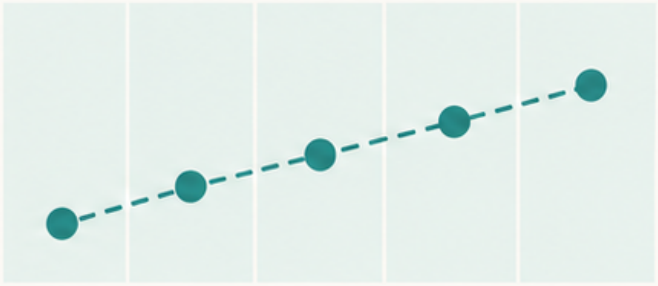
Past observed variables
Demand, observed temperature

Known future inputs
Calendar, holidays

Static features
Building type, location

TFT
Recurrence + attention

Multi-horizon forecast



Handles heterogeneous inputs

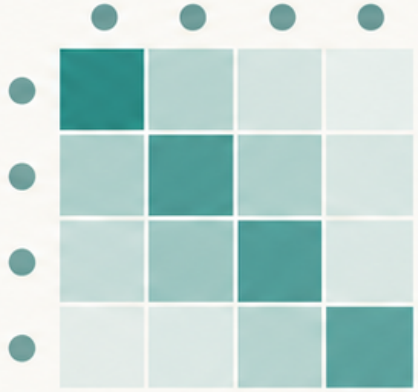
Interpretable variable importance

Designed for multi-horizon forecasting

Lim et al., arXiv:1912.09363

Are Transformers Always Better?

Transformer models



Not necessarily.

Simple baselines

Linear and seasonal models

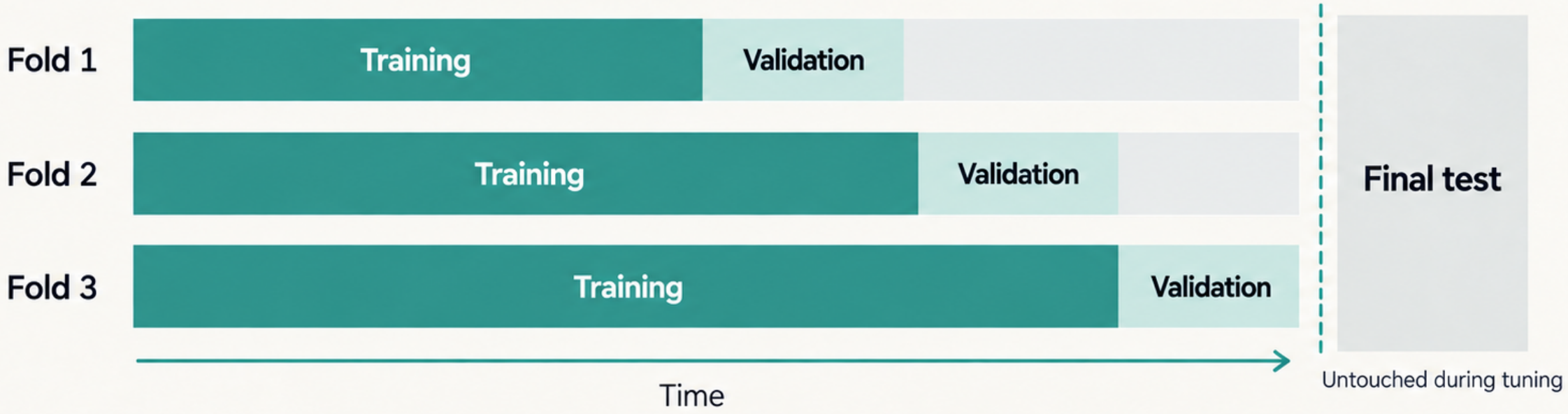
Simple linear models beat several Transformer baselines in the studied benchmarks.

- Performance depends on the data
- Simple baselines can be strong
- Use identical horizons and splits

Zeng et al., arXiv:2205.13504

Evaluation Still Matters

Walk-forward validation



Prevent leakage
Fit preprocessing on training data

Compare temporal baselines
Naïve and seasonal naïve

Choose suitable metrics
MAE · RMSE · MAPE
MAPE requires nonzero, ratio-scale targets

Source: Forecasting: Principles and Practice (3rd ed.)

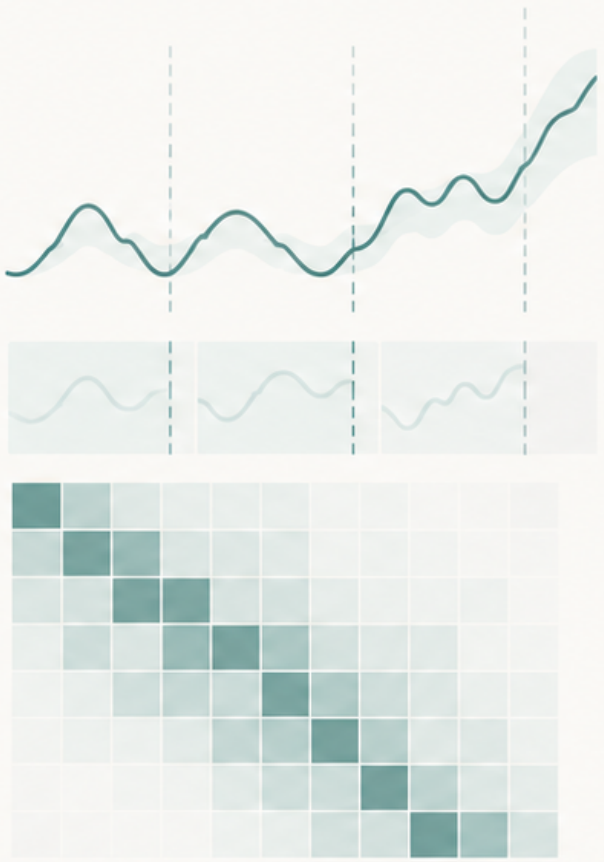
Key Takeaways

01 Transformers model sequences through **attention**

02 Time-series inputs can use **windows or patches**

03 **PatchTST and TFT** show different adaptations

04 Reliable forecasting needs **validation**, baselines, and leakage prevention



References and Questions

Vaswani et al. (2017)
Attention Is All You Need
arXiv:1706.03762

Nie et al. (2023)
A Time Series is Worth 64 Words: Long-term Forecasting with Transformers
PatchTST / arXiv:2211.14730

Lim et al. (2021)
Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting
TFT / arXiv:1912.09363

Zeng et al. (2023)
Are Transformers Effective for Time Series Forecasting?
arXiv:2205.13504

Questions?